Technical Report On: Comparative Study of Traditional Machine Learning and Transformer Models for Spam Classification

Dataset: Email Spam Classification (Category and Message columns)

Models: Logistic Regression, Naive Bayes, and DistilBERT

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1.Abstract:

Spam message classification is an important natural language processing task that aims to distinguish unwanted messages from legitimate messages. This project compares two traditional machine learning algorithms, Logistic Regression and Naive Bayes, with a transformer-based deep learning model, DistilBERT, for binary spam classification.
The traditional models use Term Frequency–Inverse Document Frequency (TF-IDF) to represent text messages as numerical features. DistilBERT uses a pretrained language model and is fine-tuned using PyTorch with a custom training loop. The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Their computational costs are also compared using training time, and the DistilBERT training loss is examined to understand its training behaviour.

The results show that DistilBERT achieved the highest accuracy of approximately 99.01%, compared with 96.68% for Logistic Regression and 96.95% for Naive Bayes. However, DistilBERT required substantially more training time. The findings demonstrate the trade-off between classification performance and computational efficiency.

2.Objectives:

The main objectives of this project are:

- To develop a spam message classification system using traditional machine learning algorithms.
- To implement DistilBERT for text classification using PyTorch and a custom training loop.
- To evaluate all three models using the same classification metrics.
- To compare model performance, training behaviour, and computational cost.
- To identify the advantages and limitations of traditional machine learning and transformer-based approaches.

3.Dataset:

The project uses a labelled email/message dataset containing two columns: "Category" and "Message". The "Category" column identifies whether a message is classified as ham or spam, while the "Message" column contains the message text.
This is a supervised binary classification problem. Ham represents legitimate messages, whereas spam represents unwanted messages. The objective is to train models that can learn from labelled examples and predict the category of an unseen message.

4.Problem Definition:

Email spam is a common problem that affects users by filling inboxes with unwanted and potentially harmful messages. Manually identifying spam emails can be time-consuming. Therefore, this project aims to develop an automated text-classification system that can distinguish between legitimate emails and spam messages.
The project compares three classification models: Logistic Regression, Naive Bayes and

DistilBERT. Logistic Regression and Naive Bayes represent traditional machine-learning approaches, while DistilBERT represents a transformer-based deep-learning approach. Their performance is evaluated using Accuracy, Precision, Recall, F1-score and Confusion Matrices.

6.Data Preparation:

The message text and category labels were prepared for use in the classification models. For the traditional machine learning models, the text messages were converted into numerical representations using TF-IDF. This representation gives greater importance to words that are informative within a document while reducing the relative importance of words that appear frequently across documents.

For DistilBERT, the messages were tokenized using the tokenizer associated with the pretrained "distilbert-base-uncased" model. Tokenization converts text into tokens and corresponding input IDs that the transformer can process. The token sequences were prepared using a maximum sequence length of 128 tokens.
The labels were represented numerically for model training, with ham assigned to class 0 and spam assigned to class 1. The data was then used for training and evaluation according to the train–test split implemented in the notebook.

7.Methodology:

7.1 Logistic Regression:

Logistic Regression was used as a traditional machine learning baseline. It learns a relationship between the numerical text features and the target classes and estimates the probability that a message belongs to a particular class. The predicted class is selected based on the model's decision threshold.
TF-IDF features were used as the input representation. Logistic Regression provides a useful baseline because it can handle sparse text features and is relatively computationally efficient.

7.2 Naive Bayes:

Naive Bayes was used as a second traditional classification algorithm. It applies Bayes' theorem to estimate the probability of a class given the observed features. The algorithm makes a conditional independence assumption about the features.
Naive Bayes is commonly used for text classification because it can learn quickly from word-based features and provides a useful comparison with Logistic Regression.

7.3 DistilBERT

DistilBERT is a transformer-based language model derived from BERT. Unlike the traditional models, which use TF-IDF features, DistilBERT processes tokenized text and uses contextual representations to capture relationships between words.
The pretrained "distilbert-base-uncased" model was adapted for binary classification. PyTorch was used to implement the training process through a custom training loop rather than the Hugging Face Trainer API. The loop performs forward propagation, calculates the loss,

computes gradients through backpropagation, and updates the trainable parameters using an optimizer.

The model was trained using a learning rate of 2×10^{-5}, a batch size of 8, and a maximum sequence length of 128 tokens. To reduce computational requirements, the first four transformer layers were frozen, while the remaining two transformer layers and classification head were fine-tuned. The initial training configuration used one epoch on the available CPU environment.

8.Design Decisions:

- Three classification models were used in this project: Logistic Regression, Naive Bayes, and DistilBERT. Logistic Regression and Naive Bayes were selected as traditional machine learning approaches, while DistilBERT was used as a transformer-based model for comparison.
- TF-IDF was used to convert text messages into numerical features for Logistic Regression and Naive Bayes. For DistilBERT, the pretrained "distilbert-base-uncased" model and its corresponding tokenizer were used.
- DistilBERT was implemented using PyTorch with a custom training loop instead of the Hugging Face Trainer API. To reduce the computational requirements of training, the first four transformer layers were frozen, while the remaining two transformer layers and the classification head were fine-tuned. A maximum sequence length of 128 tokens and a batch size of 8 were used, with a learning rate of (2 \times 10^{-5}).

9.Experimental Setup:

The experiments were carried out in a Jupyter Notebook using the email.csv dataset, which contains the Category and Message columns. The classification task involved predicting whether each message was ham or spam. The labels were represented numerically, with ham assigned to class 0 and spam assigned to class 1.

Logistic Regression and Naive Bayes were trained using TF-IDF features. DistilBERT was fine-tuned using PyTorch in the available CPU environment, with the initial configuration using one training epoch.

All three models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Their training times were also recorded for computational-cost comparison. In addition, the DistilBERT training loss was recorded and plotted to examine its training behaviour. The resulting metrics and visualizations were used to compare classification performance and computational efficiency across the three models.

9.Results And Comparative Discussion:

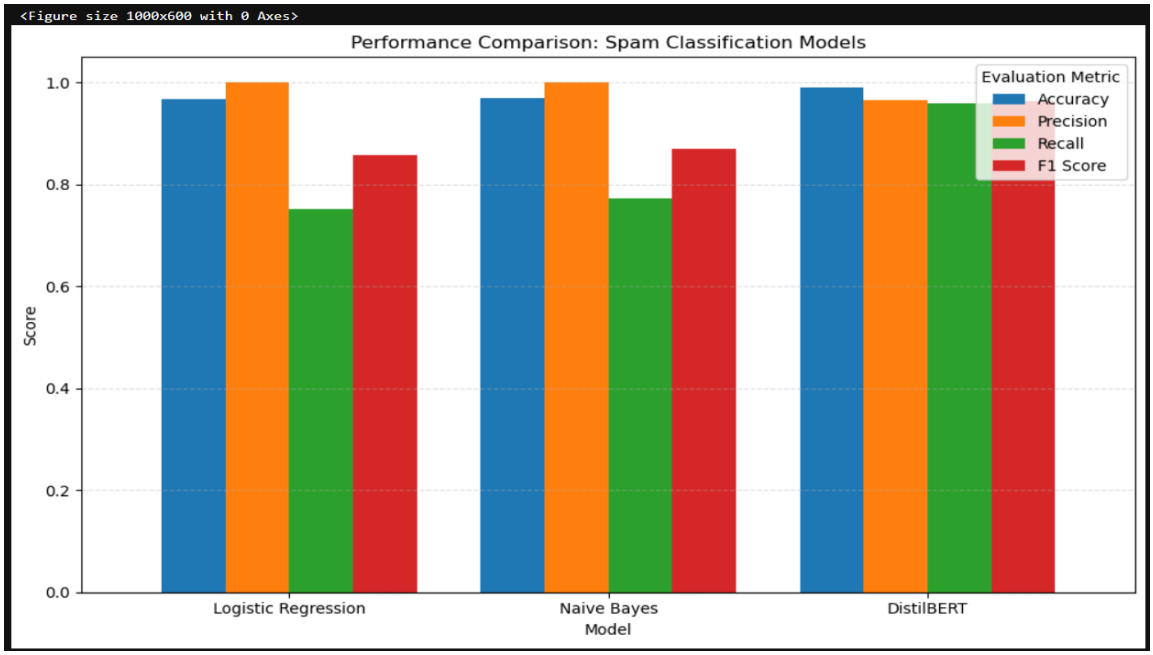
9.1 Classification Performance:

The observed accuracy results are presented in Table 1.

Table 1. Accuracy comparison of the classification models

MODEL	ACCURACY
Logistic Regression	99.68%
Naïve Bayes	96.95%
DistilBERT	99.01%

- Naive Bayes achieved slightly higher accuracy than Logistic Regression. Their similar results indicate that both traditional approaches can classify spam messages effectively using TF-IDF features.
- DistilBERT achieved the highest accuracy, at approximately 99.01%. This improvement may be due to its pretrained contextual representations, which allow it to model relationships between words rather than relying only on independent numerical text features.
- However, accuracy alone does not fully describe classification performance. Precision, recall, and F1-score must also be considered, particularly because incorrectly classifying legitimate messages as spam and failing to detect spam have different consequences.



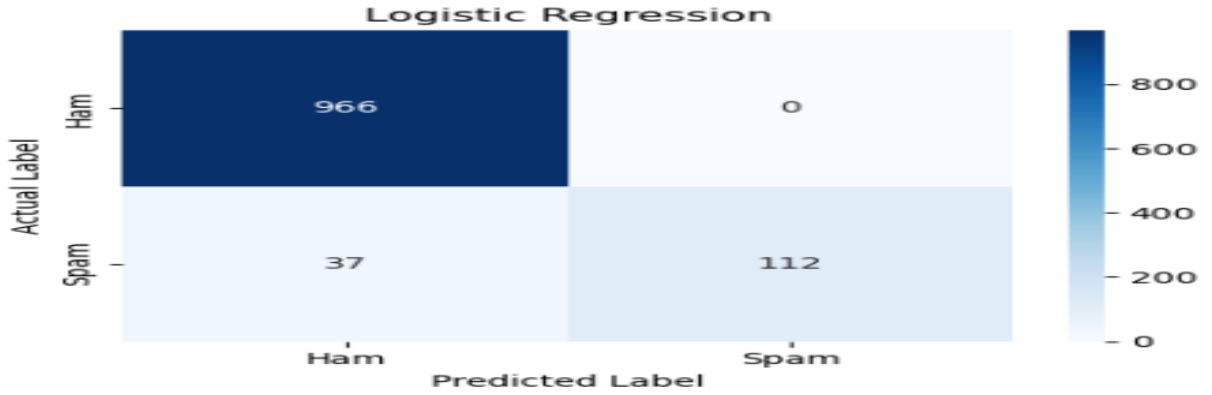
9.2 Confusion Matrix Analysis:

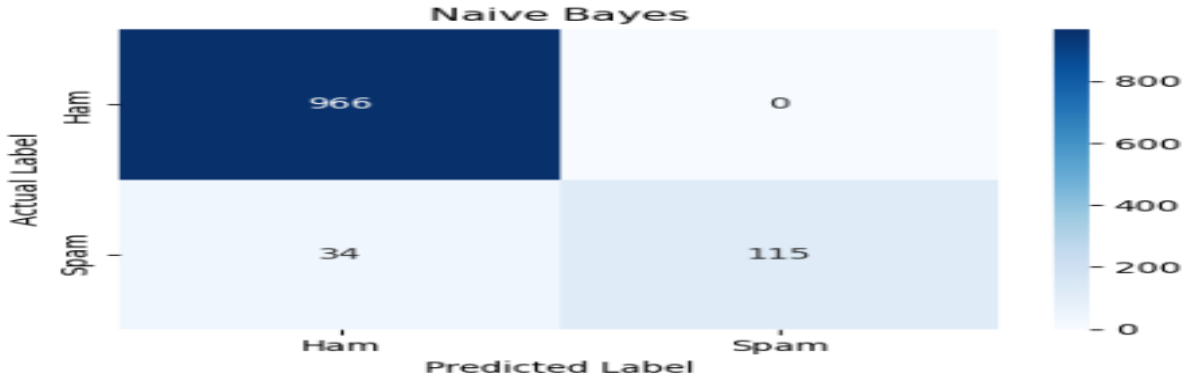
The confusion matrices provide further information about the predictions made by each model. They show the number of correctly classified ham and spam messages, together with the messages assigned to the wrong class.

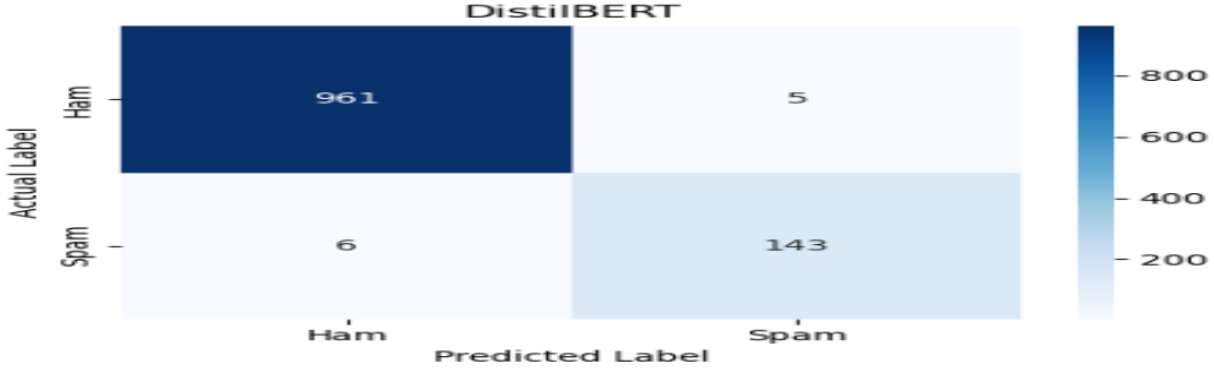
- For Logistic Regression and Naive Bayes, the confusion matrices help identify

the types of errors made by the traditional models. These results can be compared with DistilBERT to assess whether its higher accuracy is accompanied by fewer false positives and false negatives.

- In the DistilBERT confusion matrix, 1,104 out of 1,115 test messages were classified correctly. The matrix showed five ham messages incorrectly predicted as spam and six spam messages incorrectly predicted as ham. This corresponds to an accuracy of approximately 99.01%.The results indicate that DistilBERT made relatively few classification errors on the test set. Nevertheless, the confusion matrices should be considered alongside precision, recall, and F1-score to understand the models' performance on each class.



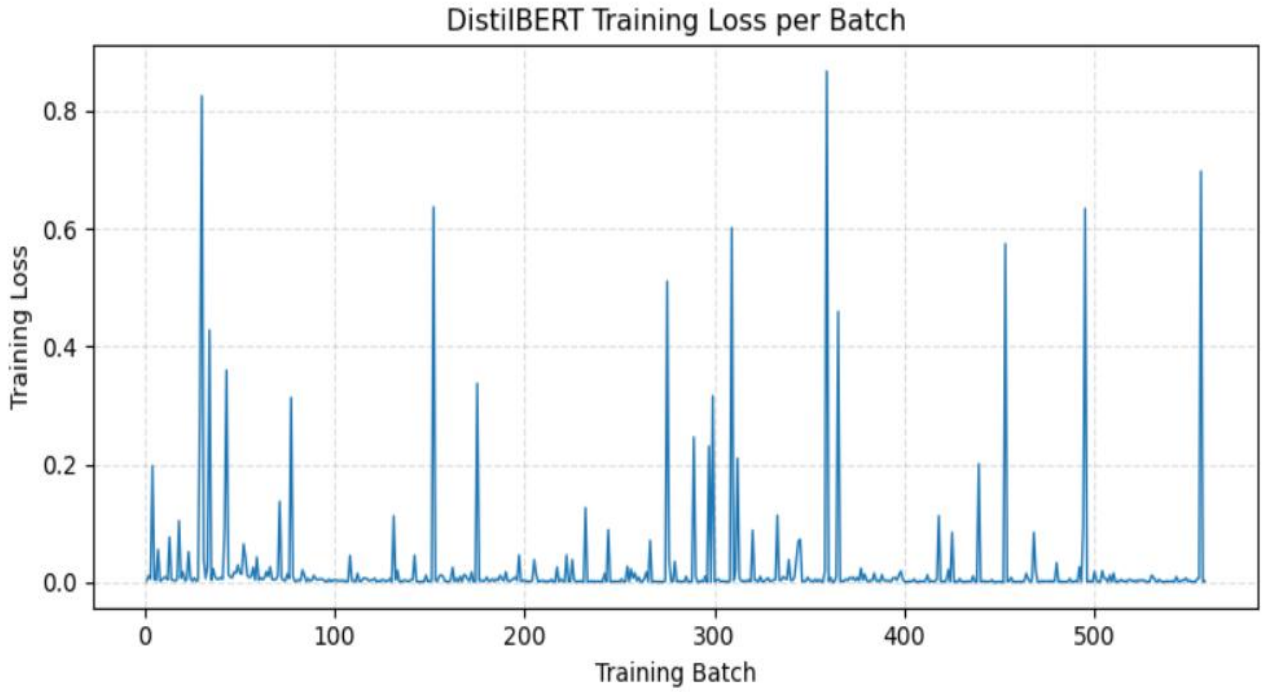




9.3 DistilBERT Training Behaviour:

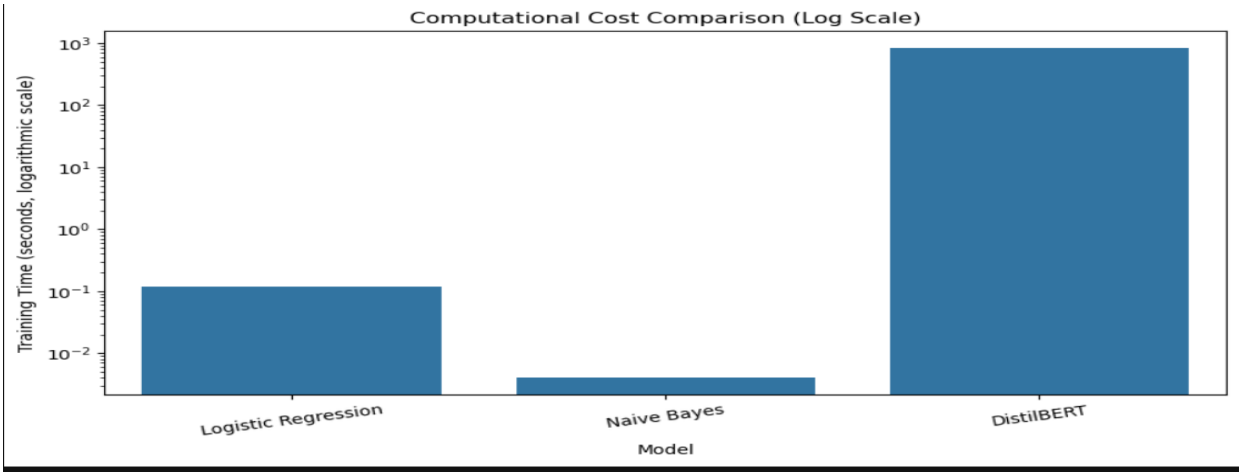
The DistilBERT training loss was recorded during training to observe how the model's error changed across training batches. The loss curve provides information about the model's learning behaviour during fine-tuning.

A decreasing training loss generally indicates that the model is improving its fit to the training data. However, training loss alone does not establish how well a model generalizes to unseen messages. Validation loss would provide additional information about possible overfitting.



9.4 Computational Cost:

- Training time was compared to evaluate the computational efficiency of the three models. The recorded times were approximately 0.12 seconds for Logistic Regression, 0.00 seconds for Naive Bayes as displayed after rounding, and 846.44 seconds for DistilBERT.
- The traditional models required substantially less time to train than DistilBERT. The displayed Naive Bayes time of 0.00 seconds should be interpreted as a rounded value rather than proof that training required no time.
- DistilBERT required considerably more computation because it performs transformer-based processing and fine-tuning. Although it achieved the highest accuracy, its training cost was much greater. Therefore, the choice of model depends on whether classification performance or computational efficiency is the main priority.



	Model	Training Time (seconds)
0	Logistic Regression	0.12
1	Naive Bayes	0.00
2	DistilBERT	846.44

Overall, DistilBERT achieved the best observed accuracy, while Logistic Regression and Naive Bayes offered faster and less computationally demanding alternatives. The traditional models are suitable when resources and training time are limited. DistilBERT may be preferable when improved classification performance justifies the additional computational cost.

10.Limitations:

- First, the experiments were conducted on a particular labelled dataset. Performance on other datasets, message types, or changing spam patterns may differ. The reported results should therefore not be assumed to represent performance in every real-world setting.
- Second, DistilBERT required substantially more training time than the traditional models in the available CPU environment. This increases the computational cost of fine-tuning and may make the transformer approach less practical when resources are limited.
- Third, the DistilBERT configuration used a maximum sequence length of 128 tokens and froze the first four transformer layers. These choices reduced the training workload, but alternative configurations could produce different results.
- Fourth, the initial training configuration used one epoch. Additional training, systematic hyperparameter tuning, and validation-based model selection could provide further insight into performance. The training loss alone is not sufficient to rule out overfitting.
- Finally, accuracy and other test metrics reflect performance on the selected test set. A larger or more diverse evaluation dataset would help establish how reliably the models perform on unseen messages.

11.Conclusion:

This project compared Logistic Regression, Naive Bayes, and DistilBERT for spam message classification. The traditional models used TF-IDF features, whereas DistilBERT used contextual text representations and was fine-tuned using PyTorch with a custom training loop.

DistilBERT achieved the highest observed accuracy, at approximately 99.01%, compared with 96.68% for Logistic Regression and 96.95% for Naive Bayes. However, its training time was considerably higher. Logistic Regression and Naive Bayes were more computationally efficient, making them useful alternatives when fast training and lower resource requirements are important.

In conclusion, the experiments demonstrate a trade-off between predictive performance and computational cost. DistilBERT provided the strongest accuracy in this experiment, while the traditional models offered faster and simpler approaches. Future work could investigate additional epochs, hyperparameter tuning, validation-based evaluation, and testing on a broader range of messages.